(a) Absolute error across the 15-model NEB campaign

Model used in NEB main figure	in-gap fixed	gap -> QL fixed	within QL fixed	in-gap self	gap -> QL self	within QL self	pristine sliding	Cr-in-gap sliding
MP-0B3	0.38	0.035	23	1.1	0.035	1.8	0.6	0.64
OMAT-0	0.31	0.027	22	0.97	0.027	NC	0.28	0.66
MPA0	0.35	0.055	25	1	0.055	1.8	0.41	0.59
FT-600K / s123	0.035	0	9.6	1.3	0	1.8	0.88	0.7
FT-600K / s234	0.45	0	12	1.3	0	1.8	0.21	0.32
FT-600K / s345	0.32	0	11	1.4	0	1.8	0.4	0.22
FT-MultiT / s123	0.37	0.066	11	1.4	0.066	1.8	0.82	0.78
FT-MultiT / s234	0.36	0.0043	10	1.4	0.0017	NC	0.23	0.033
FT-MultiT / s345	0.36	0	11	1.5	0	NC	1.1	0.97
Scratch-600K / s123	0.6	0.07	0.4	0.98	0.07	1.8	0.76	1.6
Scratch-600K / s234	0.28	0	0.44	1.2	0	1.8	0.87	2.5
Scratch-600K / s345	0.43	0	0.5	1.1	0	1.8	0.28	0.37
Scratch-MultiT / s123	0.6	0	1.2	0.84	0	1.5	0.32	1.2
Scratch-MultiT / s234	1.1	0	0.57	1.1	0	1.7	0.4	1.3
Scratch-MultiT / s345	0.67	0	0.48	1	0	1.8	0.013	0.42

NEB pathway or sliding system

Absolute error (eV)

Foundation · Fine-tuning · Scratch

(b) Fixed-path and self-relaxed migration error

Absolute barrier error (eV)

Migration pathway: in-gap · gap -> QL · within QL

Foundation · Fine-tuning · Scratch · Foundation checkpoint · Training seed · Median · NC / diagnostic

(c) Within-QL barrier: fixed path versus self-relaxed

Predicted within-QL barrier (eV)

Training strategy: Foundation · Fine-tuning · Scratch

Foundation · Fine-tuning · Scratch · Foundation checkpoint · Training seed · Self-relaxed · NC / diagnostic · DFT reference

(d) Sliding peak energy across the NEB model roster

Predicted sliding peak (eV)

DFT 2.571

DFT 1.654

Sliding system: Pristine · Cr-in-gap

Foundation · Fine-tuning · Scratch · Foundation checkpoint · Training seed · DFT reference